



CUBAN NATIONAL CHEMISTRY CONTEST 2019

ANSWERS TO THE PROBLEMS

"Imagination is more important than knowledge."

Albert Einstein



Ministerio de Educación
República de Cuba



Table of contents

Constants, conversions and equations.....	3
Problem 1. Miscellanea (20 ptos).....	4
Problem 2. Chemical kinetics (15 ptos).....	6
Problem 3. Stoichiometry I (15 ptos).....	9
Problem 4. Stoichiometry II (15 ptos).....	12
Problem 5. Structure and bonding (20 ptos).....	15
Problem 6. Thermochemistry (15 ptos).....	18
Problem 7. Solutions (15 ptos).....	20
Problem 8. Ionic equilibrium (20 ptos).....	23
Problem 9. Electrochemistry (15 ptos).....	26
Problem 10. Organic chemistry I (15 ptos).....	29
Problem 11. Organic chemistry II (20 ptos).....	31

Problem distribution per examen and grade

	Day 1	Day 2
10mo	Problems 1, 2 y 3	Problems 4, 5 y 6
11no	Problems 1, 2 y 3	Problems 7, 8 y 9
12mo	Problems 1, 2 y 3	Problems 9, 10 y 11

Author

MSc. Gerardo Manuel Ojeda Carralero

Revision

MSc. Luis Enrique Guerra Castaño

CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	Thermodynamics First Principle	$Q = \Delta E + W$
Zero in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$		$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF THE ELEMENTS

1 1A																								18 8A					
1 H 1.008	2 2A																							13 3A	14 4A	15 5A	16 6A	17 7A	2 He 4.003
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18												
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95												
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80												
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3												
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)												
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)												

Problem 1. Miscellanea (20 points)

1.1	...	Total	Total
1	idem	20	20
1	idem	20	20

Select the correct answer.

1.1. Chemical element with seven electrons in “s”-type orbitals:

A) Na	B) N	C) Fr	D) Cu
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1.2. Originate a yellow precipitate when reaction in aqueous solution:

A) AgNO ₃ and K ₂ CrO ₄	B) KI and Pb(NO₃)₂	C) CaCl ₂ and NaOH	D) FeSO ₄ and KOH
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1.3. Element with the highest secondary ionization energy:

A) Be	B) Mg	C) Ca	D) Sr
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1.4. Angular molecule:

A) NH₃	B) BeCl ₂	C) SnCl ₂	D) C ₂ H ₂
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1.5. Diamagnetic species with bond order equal to 1:

A) O ₂	B) O₂²⁻	C) N ₂	D) O ₂ ⁻
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1.6. The central atom has sp³d hybridization:

A) SF₄	B) IF ₅	C) CF ₄	D) PF ₃
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1.7. Element with the highest electron affinity:

A) S	B) P	C) Si	D) Al
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1.8. Substance with high melting and boiling points:

A) LiCl	B) PH ₃	C) Na	D) HBr
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1.9. Largest ion:

A) K ⁺	B) Cl ⁻	C) Ca ²⁺	D) S²⁻
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1.10. Gas denser than air:

A) CO	B) CH ₄	C) H₂S	D) N ₂
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1.11. Substance with the highest boiling point:

A) H₂O	B) HF	C) NH ₃	D) HCl
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1.12. Apolar molecule:

A) IF	B) IF ₃	C) IF ₅	D) IF₇
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1.13. Glassware used to prepare solutions with exact concentration:

A) pipette	B) beaker	C) Erlenmeyer's flask	D) volumetric flask
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1.14. All the carbon atoms have sp³ hybridization:

A) 2-butene	B) ethoxyethene	C) butanone	D) ethyl ethanoate
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1.15. This species can be represented by three resonance structures in which formal charges can only take the values -1, 0, +1:

A) ClO ₄ ⁻	B) SO₃²⁻	C) HCO ₃ ⁻	D) BrO ₂ ⁻
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1.16. Substance more soluble in water:

A) H ₂ S	B) Ca(OH) ₂	C) O ₂	D) CH₃CH₂OH
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1.17. Element with green color when in the flame:

A) Ca	B) Cu	C) K	D) Pb
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1.18. Element impossible to keep in the solid-state during summer months in Cuba:

A) Sr	B) Ga	C) Na	D) Os
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1.19. Noble metal:

A) Mn	B) Rb	C) Zn	D) Pt
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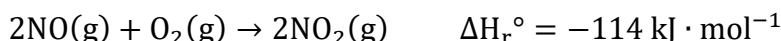
1.20. Redox reaction:

A) $\text{Na}_2\text{S}_2\text{O}_3(\text{ac}) + 2\text{HCl}(\text{ac}) = \text{S}(\text{s}) + \text{SO}_2(\text{g}) + \text{H}_2\text{O} + 2\text{NaCl}(\text{ac})$
B) $\text{CaCO}_3(\text{s}) + 2\text{HCl}(\text{ac}) = \text{CaCl}_2(\text{ac}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}$
C) $\text{NH}_3(\text{ac}) + \text{H}_2\text{SO}_4(\text{ac}) = \text{NH}_4\text{HSO}_4(\text{ac})$
D) $\text{K}_2\text{Cr}_2\text{O}_7(\text{ac}) + 2\text{KOH}(\text{ac}) = 2\text{K}_2\text{CrO}_4(\text{ac}) + \text{H}_2\text{O}$

Problem 2. Chemical kinetics (15 points)

2.1	2.2	2.3	2.4	Total	Total
12	7	3	8	30	15
12	7	3	8	30	15

When a car releases nitrogen monoxide to the atmosphere, the following reaction happens.



2.1. Find the reaction rate law and calculate the value of the specific rate constant using the kinetic data below.

Experiment	Initial concentration ($\text{mol} \cdot \text{L}^{-1}$)		$-\text{dc}(\text{NO})/\text{dt}_{\text{initial}}$ ($\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$)
	NO	O_2	
1	0,012	0,020	0,102
2	0,024	0,020	0,407
3	0,024	0,040	0,816

$$v = k \cdot c^a(\text{NO}) \cdot c^b(\text{O}_2)$$

$$v = -\frac{1}{2} \cdot \frac{\text{dc}(\text{NO})}{\text{dt}} \quad \text{for the stoichiometric relation}$$

$$\frac{\text{Exp 1}}{\text{Exp 2}} = \frac{1/2 \cdot 0,102}{1/2 \cdot 0,407} = \frac{k \cdot 0,012^a \cdot 0,020^b}{k \cdot 0,024^a \cdot 0,020^b}$$

$$0,251 = 0,5^a \rightarrow a = 2$$

3 marks

$$\frac{\text{Exp 2}}{\text{Exp 3}} = \frac{1/2 \cdot 0,407}{1/2 \cdot 0,816} = \frac{k \cdot 0,024^a \cdot 0,020^b}{k \cdot 0,024^a \cdot 0,040^b}$$

$$0,499 = 0,5^b \rightarrow b = 1$$

3 marks

$$v = k \cdot c^2(\text{NO}) \cdot c(\text{O}_2)$$

3 marks

In experiment 1

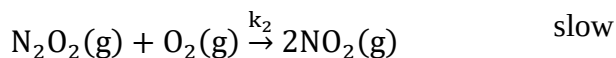
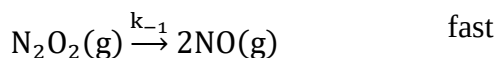
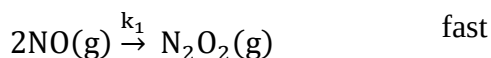
$$k = \frac{v}{c^2(\text{NO}) \cdot c(\text{O}_2)} = \frac{1/2 \cdot 0,102}{0,012^2 \cdot 0,020} = 1,77 \cdot 10^4 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$$

3 marks

It is not mandatory to calculate the average constant.

If the student does not consider factor $\frac{1}{2}$ relating the reaction rate with the rate of consumption of NO for the calculation of the rate constant, a penalty of 2 marks follows. This factor is not needed for the calculation of the partial reaction orders.

For the previous reaction, the following mechanism has been proposed.



2.2. Find an expression for the reaction rate as a function of the rate constant of the individual steps and the concentration of the reactants. Consider stationary conditions for the intermediaries.

Answer 1: The reaction rate depends fundamentally on the slow step

$$v = k_2 \cdot c(\text{N}_2\text{O}_2) \cdot c(\text{O}_2) \quad 1 \text{ mark}$$

Assuming stationary conditions for N_2O_2

$$\frac{dc(\text{N}_2\text{O}_2)}{dt} = 0 = k_1 \cdot c^2(\text{NO}) - k_{-1} \cdot c(\text{N}_2\text{O}_2) - k_2 \cdot c(\text{N}_2\text{O}_2) \cdot c(\text{O}_2) \quad 3 \text{ marks}$$

$$\frac{k_1 \cdot c^2(\text{NO})}{k_{-1} + k_2 \cdot c(\text{O}_2)} = c(\text{N}_2\text{O}_2) \quad 1 \text{ mark}$$

$$v = k_2 \cdot c(\text{N}_2\text{O}_2) \cdot c(\text{O}_2) = \frac{k_2 \cdot k_1 \cdot c^2(\text{NO}) \cdot c(\text{O}_2)}{k_{-1} + k_2 \cdot c(\text{O}_2)} \quad 2 \text{ marks}$$

The previous expression can be obtained starting from $v = \frac{1}{2} \cdot dc(\text{NO})/dt$.

Answer 2: Find an expression for the rate of consumption of NO

$$dc(\text{NO})/dt = 2k_{-1} \cdot c(\text{N}_2\text{O}_2) - 2k_1 \cdot c^2(\text{NO}) \quad 1 \text{ mark}$$

$$\frac{dc(\text{N}_2\text{O}_2)}{dt} = 0 = k_1 \cdot c^2(\text{NO}) - k_{-1} \cdot c(\text{N}_2\text{O}_2) - k_2 \cdot c(\text{N}_2\text{O}_2) \cdot c(\text{O}_2) \quad 3 \text{ marks}$$

$$\frac{k_1 \cdot c^2(\text{NO})}{k_{-1} + k_2 \cdot c(\text{O}_2)} = c(\text{N}_2\text{O}_2) \quad 1 \text{ mark}$$

$$\frac{dc(\text{NO})}{dt} = 2k_{-1} \cdot c(\text{N}_2\text{O}_2) - 2k_1 \cdot c^2(\text{NO}) = \frac{2k_{-1} \cdot k_1 \cdot c^2(\text{NO})}{k_{-1} + k_2 \cdot c(\text{O}_2)} - 2k_1 \cdot c^2(\text{NO})$$

$$\frac{dc(\text{NO})}{dt} = \frac{2k_{-1} \cdot k_1 \cdot c^2(\text{NO}) - 2k_1 \cdot c^2(\text{NO}) - 2k_1 \cdot c^2(\text{NO}) \cdot k_2 \cdot c(\text{O}_2)}{k_{-1} + k_2 \cdot c(\text{O}_2)}$$

$$\frac{dc(\text{NO})}{dt} = - \frac{2 \cdot k_2 \cdot k_1 \cdot c^2(\text{NO}) \cdot c(\text{O}_2)}{k_{-1} + k_2 \cdot c(\text{O}_2)} \quad 2 \text{ marks}$$

2.3. Under which experimental conditions the expression found by you in 2.2 is equivalent to the rate law from 2.1?

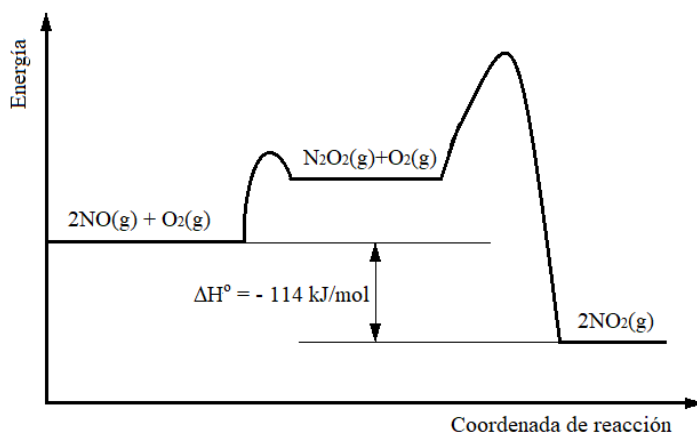
$$v = k \cdot c^2(\text{NO}) \cdot c(\text{O}_2) \text{ when } k_{-1} \ll k_2 \cdot c(\text{O}_2).$$

1 mark

This occurs if the concentration of oxygen is low.

2 marks

2.4. Draw the approximate form of an energy diagram *versus* reaction coordinate that matches the previously proposed mechanism. Indicate the enthalpy of the reaction.



8 marks

1 mark for the graph, 1 mark for reactants, 1 mark for intermediate, 1 mark for products, 1 mark for transition state 1, 1 mark for transition state 2, 1 mark activation energy 2 larger than activation energy 1, 1 mark for enthalpy change.

Problem 3. Stoichiometry I (15 points)

3.1	3.2	Total	Total
15	15	30	15
15	15	30	15

A method to determine the empirical formula of an organic compounds is to carry out the complete combustion of a sample and analyze the resultant product mixture. The final result of the combustion of organic molecules with oxygen depends on the reaction conditions. In the presence of excess oxygen, the products are carbon dioxide and water; when the oxygen content is deficient or there are other atoms different than carbon or hydrogen, other products are formed. In the following two different situations in which this method is employed are described.

I- A 4,71 g sample of a solid organic substance **A** is introduced in a previously vacuumed container of 2,00 L capacity. In the container, oxygen was added until a pressure of 390,4 kPa was reached at 25 °C. After ignition, a black solid residual with mas 0,240 g was obtained. Additionally, 11,0 g of carbon dioxide and 2,70 g of water were formed. All the initial oxygen was consumed after the reaction.

3.1. Find the empirical formula of **A** if it contained carbon, hydrogen and oxygen. Neglect the water vapor pressure and the volume of the solids.

$$n(\text{O}_2) = \frac{PV}{RT} = \frac{390,4 \cdot 2}{8,314 \cdot 298,15} = 0,315 \text{ mol} \quad 1 \text{ mark}$$

$$m(\text{O}_2) = 0,315 \cdot 32 = 10,08 \text{ g} \quad 1 \text{ mark}$$

$$m(\text{O}_2) + m(\text{A}) = 10,08 + 4,71 = 14,79 \text{ g} \quad 1 \text{ mark}$$

The products of the incomplete combustion are C, CO, CO₂ and H₂O.

$$14,79 = m(\text{C}) + m(\text{CO}) + m(\text{CO}_2) + m(\text{H}_2\text{O}) \quad \text{by the Law of conservation of mass}$$

$$m(\text{CO}) = 14,79 - m(\text{C}) - m(\text{CO}_2) - m(\text{H}_2\text{O})$$

$$m(\text{CO}) = 14,79 - 0,240 - 11,0 - 2,70 = 0,85 \text{ g} \quad 2 \text{ marks}$$

$$n(\text{CO}) = \frac{0,85}{12 + 16} = 0,0304 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{C}) = \frac{0,240}{12} = 0,0200 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{CO}_2) = \frac{11,0}{12 + 2 \cdot 16} = 0,0250 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{H}_2\text{O}) = \frac{2,70}{2 \cdot 1 + 16} = 0,0150 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{C})_{\text{A}} = n(\text{C}) + n(\text{CO}) + n(\text{CO}_2) = 0,020 + 0,0304 + 0,25 = 0,300 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{H})_{\text{A}} = 2 \cdot n(\text{H}_2\text{O}) = 2 \cdot 0,0150 = 0,300 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{O})_{\text{A}} = n(\text{H}_2\text{O}) + n(\text{CO}) + 2 \cdot n(\text{CO}_2) - 2 \cdot n(\text{O}_2)$$

$$n(\text{O})_{\text{A}} = 0,150 + 0,0304 + 2 \cdot 0,250 - 2 \cdot 0,315 = 0,0504 \text{ mol} \quad 2 \text{ marks}$$

$$n(\text{C}):n(\text{H}):n(\text{O}) = 0,300:0,300:0,0504 = 6:6:1 \rightarrow \text{C}_6\text{H}_6\text{O} \quad 2 \text{ marks}$$

- I- The complete combustion of a sample of alkanes **B** and **C** originated 61,6 g of carbon dioxide and 29,7 g of water. **B** is the higher homologous (it has an additional carbon in the molecular formula) of **C**. The molar ratio between the amounts of **B** and **C** in the mixture is 3:2, respectively.

3.2. Find the molecular formula of **B** and **C**.

As **B** is the higher homologous of **C**,

$$\text{Sean } \text{C} = \text{C}_a\text{H}_{2a+2} \text{ y } \text{B} = \text{C}_{a+1}\text{H}_{2a+4} \quad 2 \text{ marks}$$

$$\text{Combustion of C: } \text{C}_a\text{H}_{2a+2}(\text{l}) \rightarrow a\text{CO}_2(\text{g}) + (a+1)\text{H}_2\text{O}(\text{l}) \quad 1 \text{ mark}$$

$$\text{Combustion of B: } \text{C}_{a+1}\text{H}_{2a+4}(\text{l}) \rightarrow (a+1)\text{CO}_2(\text{g}) + (a+2)\text{H}_2\text{O}(\text{l}) \quad 1 \text{ mark}$$

$$n(\text{CO}_2) = \frac{61,6}{12 + 2 \cdot 16} = 1,40 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{H}_2\text{O}) = \frac{29,7}{2 \cdot 1 + 16} = 1,65 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{H}_2\text{O}) = (a+2) \cdot n(\text{B}) + (a+1) \cdot n(\text{C}) \quad 1 \text{ mark}$$

$$n(\text{CO}_2) = (a+1) \cdot n(\text{B}) + a \cdot n(\text{C}) \quad 1 \text{ mark}$$

Answer 1

$$\frac{n(\text{H}_2\text{O})}{n(\text{CO}_2)} = \frac{(a+2) \cdot n(\text{B}) + (a+1) \cdot n(\text{C})}{(a+1) \cdot n(\text{B}) + a \cdot n(\text{C})} \quad 1 \text{ mark}$$

$$\text{Substitutiong } n(\text{B}) = 3/2 \cdot n(\text{C})$$

$$\frac{1,65}{1,40} = \frac{(a+2) \cdot 1,5 \cdot n(\text{C}) + (a+1) \cdot n(\text{C})}{(a+1) \cdot 1,5 \cdot n(\text{C}) + a \cdot n(\text{C})} \quad 1 \text{ mark}$$

$$1,18 = \frac{1,5a + 3 + a + 1}{1,5a + 1,5 + a} = \frac{2,5a + 4}{2,5a + 1,5}$$

$$2,95a + 1,77 = 2,5a + 4 \rightarrow a = 4,96 \approx 5 \quad 1 \text{ mark}$$

$$\text{C} = \text{C}_5\text{H}_{12} \quad \text{B} = \text{C}_6\text{H}_{14} \quad 4 \text{ marks}$$

Answer 2

Substituting $n(B) = 3/2 \cdot n(C)$

$$\left. \begin{aligned} 1,65 &= (a + 2) \cdot 3/2 \cdot n(C) + (a + 1) \cdot n(C) \\ 1,40 &= (a + 1) \cdot 3/2 \cdot n(C) + a \cdot n(C) \end{aligned} \right\} \quad 1 \text{ mark}$$

$$\left. \begin{aligned} 1,65 &= 2,5 \cdot a \cdot n(C) + 4 \cdot n(C) \\ 1,40 &= 2,5 \cdot a \cdot n(C) + 1,5 \cdot n(C) \end{aligned} \right\}$$

Subtracting the second equation from the first, one gets

$$0,25 = 2,5 \cdot n(C) \rightarrow n(C) = 0,1 \text{ mol} \quad 1 \text{ mark}$$

Substituting in the first equation,

$$1,65 = 2,5 \cdot a \cdot 0,1 + 4 \cdot 0,1$$

$$1,65 = 0,25 \cdot a + 0,4 \rightarrow a = 5 \quad 1 \text{ mark}$$

$$C = C_5H_{12} \quad B = C_6H_{14} \quad 4 \text{ marks}$$

Problem 4. Stoichiometry II (15 points)

4.1	4.2	4.3	4.4	Total	Total
14	3	3	10	30	15
14	3	3	10	30	15

Compound derivatives from carbon allotropes fullerene and graphite catch the interest of researchers in the field of material sciences.

Buckminsterfullerene (C_{60} , the most abundant fullerene) can form salts of general formula M_nC_{60} ; where M is a metal and n is an integer number. Interest on such compounds was dramatically increased when it was discovered that many are good electric conductors in the solid state. For example, certain alkaline metal **X** combines with C_{60} in different proportions to form salts **A** and **B**, containing 14,0% and 24,6% (in mass) of metal, respectively. **A** is a conductor in the solid state and superconductor below 19,3 K, while **B** is an insulator.

4.1. Find metal **X** and the formula of **A** and **B**.

In compounds with the formula X_nC_{60} , the mass fraction of the metal is

$$\omega = \frac{n \cdot A_r(X)}{n \cdot A_r(X) + 60 \cdot 12} \quad 2 \text{ marks}$$

$$\omega \cdot n \cdot A_r(X) + \omega \cdot 60 \cdot 12 = n \cdot A_r(X)$$

$$\frac{\omega \cdot 720}{(1 - \omega) \cdot n} = A_r(X) \quad 2 \text{ marks}$$

In salt A, $\omega = 0,14$

Para $n = 1$ $A_r(X) = 117,2$ $X = ?$

Para $n = 2$ $A_r(X) = 58,6$ $X = Co$ It is not an alkaline metal.

Para $n = 3$ $A_r(X) = 58,6$ $X = K$ 2 marks

In salt B, $\omega = 0,246$

Para $n = 1$ $A_r(X) = 234,9$ $X = ?$

Para $n = 2$ $A_r(X) = 117,5$ $X = ?$

Para $n = 3$ $A_r(X) = 78,3$ $X = ?$

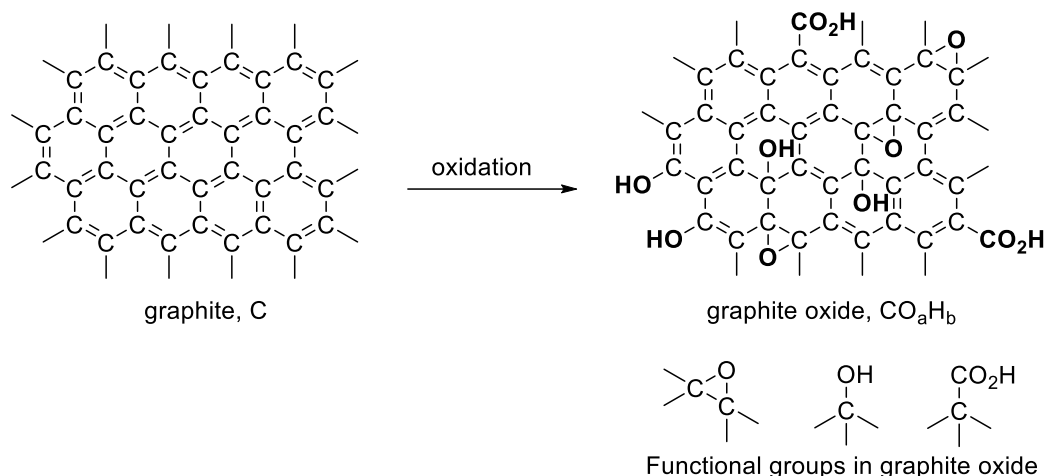
Para $n = 4$ $A_r(X) = 58,7$ $X = Co$

Para $n = 5$ $A_r(X) = 46,98$ $X = ?$

Para $n = 6$ $A_r(X) = 39,2$ $X = K$ 2 marks

$M = K$ $A = K_3C_{60}$ $B = K_6C_{60}$ 6 marks

Graphite oxide (GO) is obtained by treating graphite with strong oxidant reagents. In GO the hexagonal layers of carbon atoms contain oxygen atoms in the form of a variety of functional groups. In recent years, GO has attracted the attention as a plausible precursor of graphene, a carbon two-dimensional nanomaterial with unique electric properties. The molecular formula of GO is CO_aH_b , and the subscripts **a** and **b** depend on the oxidation method.



The simplest model for GO layers, Hoffman's model, assumes that only one functional group (-O-) is formed in the carbon layer resulting from graphite oxidation.

4.2. Calculate **a** in the formula CO_a of a GO sample in which 25% of carbon atoms are oxidized (meaning they are attached to an oxygen atom). Assume that Hoffman's model is valid.

Supposing 1 mol amount of carbon atoms, the amount of oxidized atoms is equal to the amount of C-O bonds and it is 0,25 mol. The amount of oxygen atoms will be half of the amount of C-O bonds, 0,125 mol. The formula of the GO is

$\text{CO}_{0,125}$

3 marks

4.3. Which is the maximum value that can take **a** in Hoffman's Model?

The highest value will be reached when all carbon atoms are oxidized, that is, bonded to an oxygen atom. In this case, the amount of C-O bonds will be 1 mol and the amount of oxygen atoms will be 0,5 mol. The formula of the GO is

$\text{CO}_{0,5}$

3 marks

A more updated model of a GO layer, Lerf-Klinowski's model, considers that the oxygen atoms are present in three functional groups (-O-, -OH, -COOH; see the figure).

4.4. Calculate the amount of carbon atoms (in %) that are **not** oxidized in a layer of GO with formula $\text{CO}_{0,42}\text{H}_{0,20}$. Suppose that there is an equal amount of -OH and -COOH groups.

As hydrogen atoms are only present in hydroxyl (OH) and carboxylic (CO_2H) groups, the amount of these groups is 0,10 mol each.

4 marks

The amount of oxygen atoms in groups of epoxide type is that not present in hydroxyl or carboxylic groups: $0,42 - 0,10 - 2 \cdot 0,10 = 0,12$ mol. 2 marks

Oxidized carbon atoms are those bonded to oxygen atoms: two per each epoxide and one per each hydroxyl or carboxylic group: $2 \cdot 0,12 + 0,10 + 0,10 = 0,44$ mol 2 marks

The amount of carbon atoms that are not oxidized is: 0,56 mol, 56% 2 marks

Problem 5. Structure and bonding (20 points)

5.1	5.2	5.3	5.4	5.5	5.6	5.7	5.8	5.9	5.10	Total	Total
5	3	2	2	2	3	6	7	4	6	40	20
5	3	2	2	2	3	6	7	4	6	40	20

Boron is present in nature in minerals borax, $\text{Na}_2\text{B}_4\text{O}_5(\text{OH})_4 \cdot 8\text{H}_2\text{O}$ and kernite, $\text{Na}_2\text{B}_4\text{O}_5(\text{OH})_4 \cdot 2\text{H}_2\text{O}$, from which the impure chemical element is obtained. Borax is converted in boric acid by reaction with hydrochloric acid and following dehydration originates boron oxide. The oxide is reduced with magnesium, washed with alkali and then with hydrofluoric acid. In contrast, pure boron is produced by reduction of boron tribromide with hydrogen.

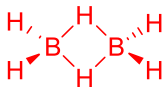
5.1. Formulate the chemical equations corresponding to the reactions previously described.



To write the oxidation states is not mandatory.

Boron tends to form four bonds in the compounds containing this atom. This explains that the simplest boron hydride, diborane (B_2H_6), exist in a dimeric form.

5.2. Represent the structure of diborane molecule.



3 marks

5.3. The reaction between diborane and ammonia generates an ionic product in which only the cation contains nitrogen atoms and both ions contain boron atoms. Suggest a formula for this compound.



The reaction between diborane and lithium hydride forms lithium tetrahydrideborate, LiBH_4 , compound used to storage hydrogen.

5.4. Write the chemical equation for the reaction between LiBH_4 and water to release hydrogen. In this process $\text{Li}[\text{B}(\text{OH})_4]$ is formed.



5.5. Boron halides react with molecules with non-bonding electron pairs, as amines, to form adducts. Why boron halides are electron deficient molecules?

Boron atom in halides has an empty p orbital that may accept the non-bonded electron pair from amines. 2 marks

5.6. The reactivity order of boron halides with amines is $\text{BF}_3 < \text{BCl}_3 \leq \text{BBr}_3$. Justify this order considering the molecular structure.

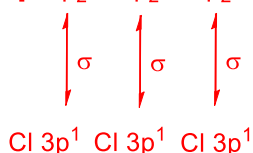
The larger the halogen atom the weaker the π interaction between the empty p orbital of boron atom and the filled p orbitals of the halogens; therefore, in the larger members of the series, the empty p orbital of boron is able to accept electrons easier. 3 marks

5.7. Write the electronic distribution of molecule BCl_3 according to the Valence Bond Theory. Consider that all the chemical bonds are equivalent.

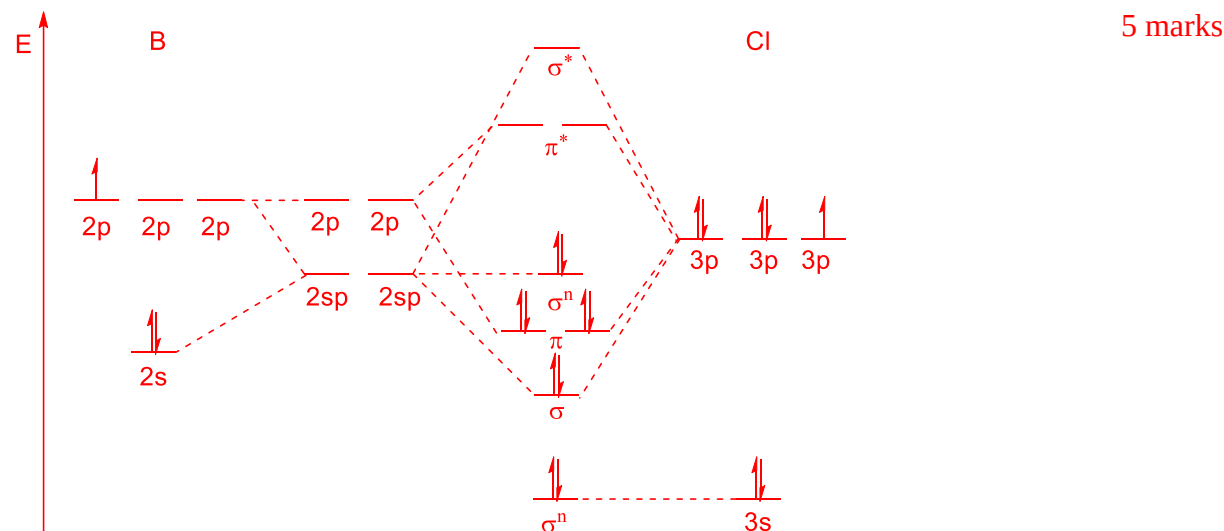
Cl: $[\text{Ar}] 3s^2 3p^2 3p^2 3p^1$ 1 mark

B: $[\text{He}] 2s^2 2p^1$ B*: $[\text{He}] 2s^1 2p^1 2p^1$ B(hib): $[\text{He}] 2sp_2^1 2sp_2^1 2sp_2^1$ 2 marks

B (hib): $[\text{He}] 2sp_2^1 2sp_2^1 2sp_2^1$ 3 marks



5.8. BCl_3 molecule decomposes by electronic impact. Use the Molecular Orbital Theory to predict the magnetic properties of product, BCl . Consider only the orbitals from the valence shell



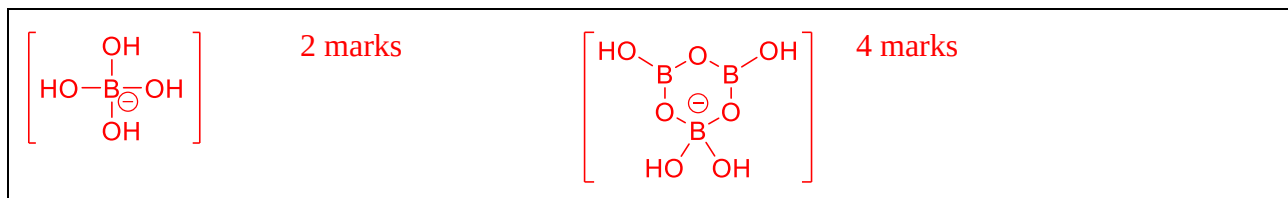
BCl molecule is diamagnetic. 2 marks

Alternative molecular orbital diagrams are accepted, for example, considering the boron atom unhybridized, as long as the orbital interactions are logic. Writing the electronic configuration instead of drawing the energy diagram is also accepted.

5.9. When boron trichloride reacts with ammonium chloride borazine, $\text{B}_3\text{N}_3\text{H}_6$, is formed. This compound is analogous to benzene. Borazine molecule is a cycle in which boron and nitrogen atoms are alternating and have sp^2 hybridization. Represent the structure of borazine.



5.10. Boric acid in aqueous solution form a variety of oxyanions depending on the acidity of the media; for example, ions $[B(OH)_4]^-$ and $[B_3O_3(OH)_4]^-$. Represent the structure of these ions.



Problem 6. Thermochemistry (15 points)

6.1	6.2	6.3	6.4	6.5	6.6	6.7	Total	Total
2	5	2	2	7	8	4	30	15
2	5	2	2	7	8	4	30	15

Hydrazine, N_2H_4 , is a fuel used in the thrusters of space crafts. It is very popular because doesn't release carbon dioxide. The fast production of hot gasses during the combustion is the source of the rocket trust.

6.1. Write the equation of the decomposition reaction of hydrazine in ammonia and nitrogen, which is a collateral reaction happening in the process.



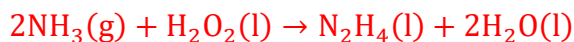
2 marks

Hydrazine can be obtained by a reaction between ammonia and hydrogen peroxide.



6.2. When hydrazine passes through and adequate catalyst, it decomposes in the parent elements. Calculate the standard enthalpy variation for this process.

$\Delta H_{\text{f}}^\circ$ ($\text{kJ} \cdot \text{mol}^{-1}$): $\text{NH}_3 = -46,1$; $\text{H}_2\text{O}_2 = -187,8$; $\text{H}_2\text{O} = -285,8$.



$$\Delta H_{\text{r}}^\circ = \Delta H_{\text{f}}^\circ(\text{N}_2\text{H}_4) + 2 \cdot \Delta H_{\text{f}}^\circ(\text{H}_2\text{O}) - 2 \cdot \Delta H_{\text{f}}^\circ(\text{NH}_3) - \Delta H_{\text{f}}^\circ(\text{H}_2\text{O}_2)$$

2 marks

$$241 = \Delta H_{\text{f}}^\circ(\text{N}_2\text{H}_4) + 2 \cdot (-285,8) - 2 \cdot (-46,1) - (-187,8)$$

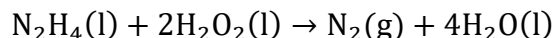
$$\Delta H_{\text{f}}^\circ(\text{N}_2\text{H}_4) = 532,6 \text{ kJ} \cdot \text{mol}^{-1}$$

1 mark



2 marks

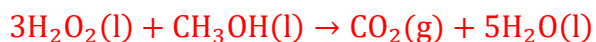
The first fighter plane thrust by a rocket, Messerschmitt Me 163, was propelled by the reaction of a mixture containing hydrazine and methanol known as "C-Stoff" and hydrogen peroxide ("T-Stoff"). Hydrogen peroxide reacts with hydrazine according to the equation



6.3. What is the oxidation number of nitrogen and oxygen in reactants and products.

N_2H_4	H_2O_2	N_2	H_2O	2 marks
-2	-1	0	-2	0,5 c/u

6.4. Hydrogen peroxide oxidizes methanol to carbon dioxide and water. Write the chemical equation corresponding to this reaction.



2 marks

6.5. A fighter plane carries 225 L of hydrazine and 862 L of methanol. Use the following data to calculate the heat released under standard conditions when all the fuel is burnt.

$\Delta H_{\text{combustión}}^{\circ}$ (kJ·mol⁻¹): N₂H₄ = -622,2; CH₃OH = -726,0;

Densidad (g·mL⁻¹): N₂H₄ = 1,021; CH₃OH = 0,7918.

$$M(\text{N}_2\text{H}_4) = 2 \cdot 14 + 4 \cdot 1 = 32 \text{ g} \cdot \text{mol}^{-1}$$

$$M(\text{CH}_3\text{OH}) = 12 + 4 \cdot 1 + 16 = 32 \text{ g} \cdot \text{mol}^{-1}$$

$$Q(\text{N}_2\text{H}_4) = \Delta H_{\text{comb}}^{\circ} \cdot \frac{\rho \cdot V}{M} = -622,2 \cdot \frac{1,021 \cdot 225000}{32} = -4,47 \cdot 10^6 \text{ kJ} \quad 3 \text{ marks}$$

$$Q(\text{CH}_3\text{OH}) = \Delta H_{\text{comb}}^{\circ} \cdot \frac{\rho \cdot V}{M} = -726,0 \cdot \frac{0,7918 \cdot 862000}{32} = -1,55 \cdot 10^7 \text{ kJ} \quad 3 \text{ marks}$$

$$Q(\text{total}) = -1,55 \cdot 10^7 + (-4,47 \cdot 10^6) = -2,00 \cdot 10^7 \text{ kJ} \quad 1 \text{ mark}$$

Besides obtention, the storage and transport of fuels possess several challenges. In this context, two central concepts are volumetric and gravimetric energy densities. The volumetric energy density is the recoverable energy divided by the total volume. The gravimetric energy density is the recoverable energy divided by the total mass.

6.6. Calculate the volumetric and gravimetric energy densities for hydrazine and methanol.

Gravimetric density energy

$$\text{N}_2\text{H}_4 \quad \frac{-622,2 \text{ kJ} \cdot \text{mol}^{-1}}{32 \text{ g} \cdot \text{mol}^{-1}} = -19,4 \text{ kJ} \cdot \text{g}^{-1} \quad 2 \text{ marks}$$

$$\text{CH}_3\text{OH} \quad \frac{-726,0 \text{ kJ} \cdot \text{mol}^{-1}}{32 \text{ g} \cdot \text{mol}^{-1}} = -22,7 \text{ kJ} \cdot \text{g}^{-1} \quad 2 \text{ marks}$$

Volumetric density energy

$$\text{N}_2\text{H}_4 \quad -19,4 \text{ kJ} \cdot \text{g}^{-1} \cdot 1,021 \text{ g} \cdot \text{mL}^{-1} = -19,8 \text{ kJ} \cdot \text{mL}^{-1} \quad 2 \text{ marks}$$

$$\text{CH}_3\text{OH} \quad -22,7 \text{ kJ} \cdot \text{g}^{-1} \cdot 0,7918 \text{ g} \cdot \text{mL}^{-1} = -18,0 \text{ kJ} \cdot \text{mL}^{-1} \quad 2 \text{ marks}$$

6.7. Considering the previous result, choose which of the two substances is more adequate to be employed as fuel.

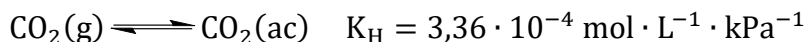
When the limiting factor is the weight, methanol is the best fuel. 2 marks

When the limiting factor is the volume, hydrazine is the best fuel. 2 marks

Problem 7. Solutions (15 points)

7.1	7.2	7.3	7.4	7.5	7.6	7.7	7.8	Total	Total
5	2	2	4	5	3	5	4	30	15
5	2	2	4	5	3	5	4	30	15

Henry's constant, K_H , relates the molar concentration with the partial pressure of the gas saturating the solvent at a given temperature. For example, at 25 °C for carbon dioxide,



The solubility of gasses can also be expressed in terms of the solubility coefficient α , defined as the volume of the gas (in milliliters) measured at standard conditions (0 °C and 101,3 kPa) that dissolves in 1 mL of water when the partial pressure of the gas is 101,3 kPa.

7.1. Find an approximate expression relating K_H and α . Consider that the volume of the solution does not depend on the amount of dissolved gas.

Answer 1

$$\alpha = \frac{V(\text{g})_{\text{TPN}}}{V(\text{d})} = \frac{n \cdot V_M(\text{TPN})}{V(\text{d})} \approx \frac{n \cdot V_M(\text{TPN})}{V(\text{D})} = \frac{n \cdot V_M(\text{TPN}) \cdot p}{V(\text{D}) \cdot p} = K_H \cdot 22,4 \cdot 101,3$$

Answer 2

$$K_H = \frac{c}{p} = \frac{n}{V(\text{D}) \cdot p} = \frac{V(\text{g})_{\text{TPN}}}{V_M(\text{TPN}) \cdot V(\text{D}) \cdot p} \approx \frac{V(\text{g})_{\text{TPN}}}{V_M(\text{TPN}) \cdot V(\text{d}) \cdot p} = \frac{\alpha}{22,4 \cdot 101,3}$$

1 mark per each step in the demonstration

5 marks

7.2. How do you explain the difference between the solubilities of gases SO_2 , H_2S and O_2 ?

α (at 25 °C): $\text{SO}_2 = 32,79$; $\text{H}_2\text{S} = 2,282$; $\text{O}_2 = 0,02831$.

The solubility is higher when increasing the polarity of the gas molecules. Besides, SO_2 , can originate hydrogen bonds. 2 marks

7.3. Why the solubility of gasses decreases when the temperature raises?

Because for gases the enthalpy change of solution is negative.

2 marks

Hydrogen chloride is a gas very soluble in water. At temperatures below 0 °C, it is possible to obtain hydrates of well-defined composition $\text{HCl} \cdot n\text{H}_2\text{O}$, where n is an integer number.

7.4. Find the formula of a hydrogen chloride hydrate that melts at -70 °C knowing that it contains 9,1% mass of hydrogen.

In compounds with formula, $\text{HCl} \cdot n\text{H}_2\text{O}$

$$M(\text{HCl}) = 1,01 + 35,45 = 36,46 \text{ g} \cdot \text{mol}^{-1}$$

$$M(\text{H}_2\text{O}) = 2 \cdot 1,01 + 16 = 18,02 \text{ g} \cdot \text{mol}^{-1}$$

$$\omega = \frac{1,01 \cdot (1 + 2n)}{36,46 + n \cdot 18,02} \quad 2 \text{ marks}$$

For $\omega = 0,091$, then $n = 6$. $\text{HCl} \cdot 6\text{H}_2\text{O}$ 2 marks

For solid substances, the solubility coefficient S is commonly used. It is defined as the solute mass that dissolves in 100 g of water. In addition, for sparingly soluble ionic salutes there is a direct relation between the coefficient S and the solubility product K_{ps} .

7.5. Demonstrate that for PbCl_2 the following relation exists, between the solubility coefficient S and the constant K_{ps} . Consider that the density of pure water is $1,00 \text{ g} \cdot \text{mL}^{-1}$ at 25°C .

$$K_{ps} \approx 4000 \cdot \frac{S^3}{M^3}$$

$$[\text{Pb}^{2+}] \approx \frac{S}{M(\text{PbCl}_2) \cdot 0,1} \quad 2 \text{ marks}$$

$$K_{ps} = [\text{Pb}^{2+}][\text{Cl}^-]^2 = [\text{Pb}^{2+}](2[\text{Pb}^{2+}])^2 \quad 1 \text{ mark}$$

$$K_{ps} = 4 \left(\frac{S}{M(\text{PbCl}_2) \cdot 0,1} \right)^3 \quad 1 \text{ mark}$$

$$K_{ps} = 4000 \cdot \frac{S^3}{M^3} \quad 1 \text{ mark}$$

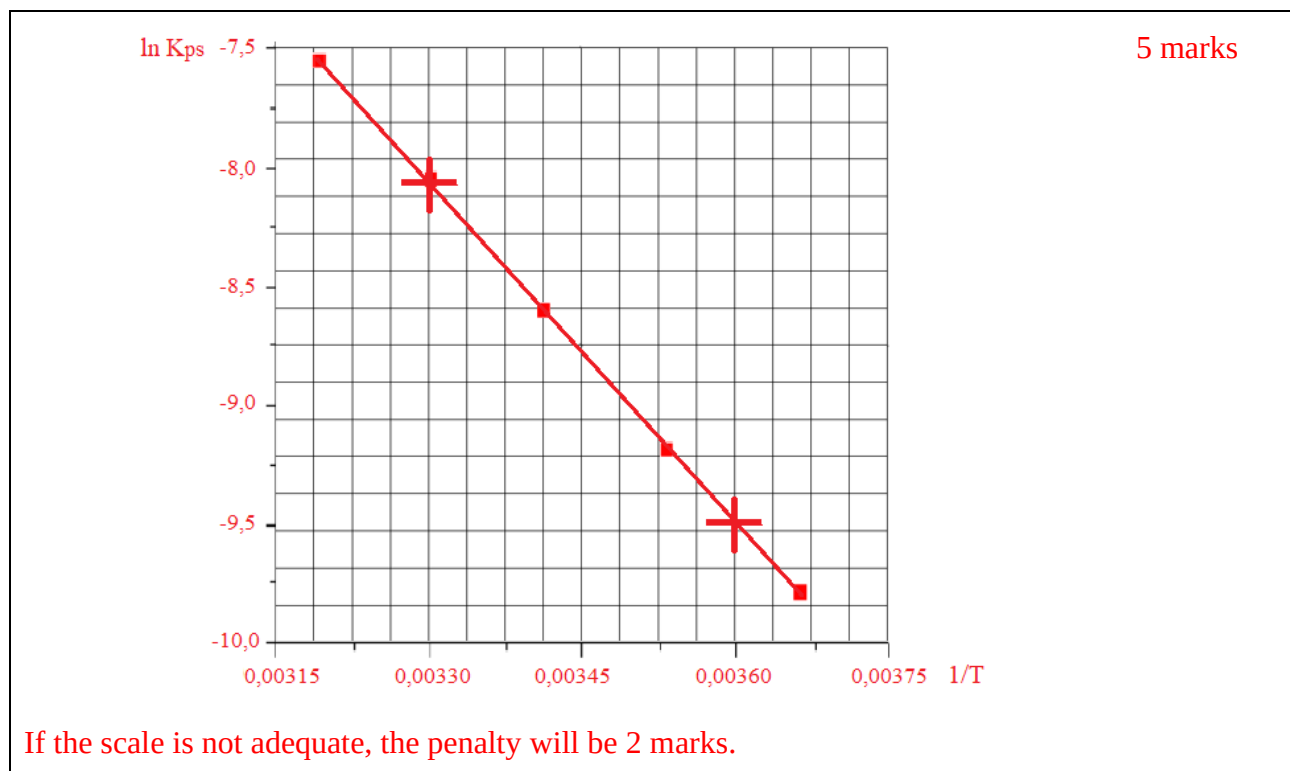
A student wanted to indirectly calculate the standard enthalpy variation of the dissolution of lead(II) chloride. To attain this objective, the student determined the solubility coefficient S of PbCl_2 at several temperatures. The results obtained by the student are shown in the following.

T ($^\circ \text{C}$)	0	10	20	30	40
1/T (K^{-1})					
S	0,67	0,82	1,00	1,20	1,42
K_{ps}					
$\ln K_{ps}$					

7.6. Complete the missing information in the previous table. Suppose that the expression from item 7.5 is valid for all the temperatures. $M(\text{PbCl}_2) = 278 \text{ g} \cdot \text{mL}^{-1}$.

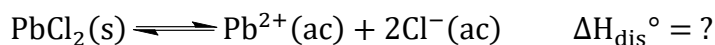
T ($^\circ \text{C}$)	0	10	20	30	40	
1/T (K^{-1})	0,00366	0,00353	0,00341	0,00330	0,00319	1 mark
S	0,67	0,82	1,00	1,20	1,42	
K_{ps}	$5,60 \cdot 10^{-5}$	$1,03 \cdot 10^{-4}$	$1,86 \cdot 10^{-4}$	$3,22 \cdot 10^{-4}$	$5,33 \cdot 10^{-4}$	1 mark
$\ln K_{ps}$	-9,79	-9,18	-8,59	-8,04	-7,54	1 mark

7.7. Draw a plot of $\ln K_{ps}$ vs $1/T$ in the indicated space below. Use an adequate scale.



7.8. Calculate the value of the standard enthalpy variation for the dissolution of PbCl_2 . Use the plot elaborated by you in item 7.7 and the following relation.

$$\ln K_{ps} = -\frac{\Delta H_{\text{dis}}^{\circ}}{R} \cdot \frac{1}{T} + \frac{\Delta S_{\text{dis}}^{\circ}}{R}$$



Employing the points highlighted in the graphic for calculating the slope,

$$-\frac{\Delta H_{\text{dis}}^{\circ}}{R} \approx \frac{-8,1 - (-9,5)}{0,00330 - 0,00360} = -4,7 \cdot 10^3 \quad 2 \text{ marks}$$

$$\Delta H_{\text{dis}}^{\circ} = -8,314 \cdot (-4,7 \cdot 10^3) = 3,9 \cdot 10^4 \text{ J} \cdot \text{mol}^{-1} = 39 \text{ kJ} \cdot \text{mol}^{-1} \quad 2 \text{ marks}$$

Values in the range $30\text{--}50 \text{ kJ} \cdot \text{mol}^{-1}$ are considered correct. Answers using analytic methods and with scientific calculators are accepted; penalty of 2 marks for not employing the graphic.

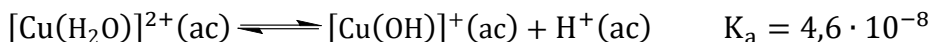
For example, by minimum square one gets

$$r = -0,9999, B = -4,588 \cdot 10^3, \Delta H_{\text{dis}}^{\circ} = 38,1 \text{ kJ} \cdot \text{mol}^{-1}$$

Problem 8. Ionic equilibrium (20 points)

8.1	8.2	8.3	8.4	8.5	8.6	8.7	8.8	Total	Total
3	2	4	6	3	8	2	12	40	20
3	2	4	6	3	8	2	12	40	20

The solutions of salts containing copper(II) ions are slightly acid, because this ion is hydrated and behaves as a weak acid in its hydrolysis. The process can be simplified to



8.1. Calculate the pH of an aqueous solution of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ $0,50 \text{ mol} \cdot \text{L}^{-1}$.

$$[\text{H}^{+}] \approx \sqrt{K_a \cdot c(\text{Cu}^{2+})} = 8,314 \cdot (-5,0 \cdot 10^3) = 4,2 \cdot 10^4 \text{ J} \cdot \text{mol}^{-1} = 42 \text{ J} \cdot \text{mol}^{-1}$$

$$[\text{H}^{+}] = \sqrt{4,6 \cdot 10^{-8} \cdot 0,5} = 1,5 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1} \quad 2 \text{ marks}$$

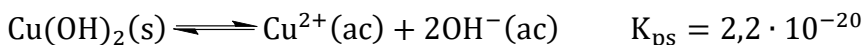
$$\text{pH} = \log[\text{H}^{+}] = \log(1,5 \cdot 10^{-4}) = 3,8 \quad 1 \text{ mark}$$

When sodium hydroxide is added to a solution of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, the formation of a light blue colored solid characteristic of copper(II) hydroxide.

8.2. Which condition is necessary for a precipitate to form?

The ionic product must be larger than the solubility product constant. 2 marks

8.3. Calculate the solubility (in $\text{mol} \cdot \text{L}^{-1}$) of copper(II) hydroxide in pure water.



$$K_{\text{ps}} = [\text{Cu}^{2+}][\text{OH}^{-}]^2 = [\text{Cu}^{2+}](2[\text{Cu}^{2+}])^2 = 4[\text{Cu}^{2+}]^3 \quad 2 \text{ marks}$$

$$[\text{Cu}^{2+}] = \sqrt[3]{\frac{K_{\text{ps}}}{4}} = \sqrt[3]{\frac{2,2 \cdot 10^{-20}}{4}} = 1,8 \cdot 10^{-7} \text{ mol} \cdot \text{L}^{-1} \quad 2 \text{ marks}$$

8.4. 25 mL of an aqueous solution of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ $0,50 \text{ mol} \cdot \text{L}^{-1}$ were mixed with 25 mL of an aqueous solution of NaOH $1,0 \text{ mol} \cdot \text{L}^{-1}$. Calculate the concentration of ions Cu^{2+} present in the resultant solution when the equilibrium is reached.



$$n(\text{Cu}^{2+}) = 25 \text{ ml} \cdot 0,50 \text{ mol/L} = 12,5 \text{ mmol} \quad 1 \text{ mark}$$

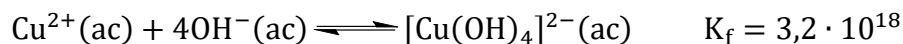
$$n(\text{OH}^{-}) = 25 \text{ ml} \cdot 1,0 \text{ mol/L} = 25 \text{ mmol} \quad 1 \text{ mark}$$

As the amounts are stoichiometric the result is equal to the previous. 2 marks

$$[\text{Cu}^{2+}] = 1,8 \cdot 10^{-7} \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

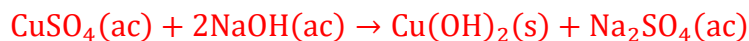
If the addition of strong base continues, the precipitate dissolves originating a marine-blue colored solution, due to the formation of the ion tetrahydroxidecuprate(II), $[\text{Cu}(\text{OH})_4]^{-2}$.

8.5. Given the value of the formation constant K_f of tetrahydroxidecuprate(II) from ions copper(II) and hydroxide, calculate the value of the equilibrium constant K between the hydroxide and complex ions.



$$K = K_{ps} \cdot K_f = 2,2 \cdot 10^{-20} \cdot 3,2 \cdot 10^{18} = 7,04 \cdot 10^{-2} \approx 0,070$$

8.6. 25 mL of an aqueous solution of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ $0,50 \text{ mol} \cdot \text{L}^{-1}$ were mixed with 75 mL of an aqueous solution of NaOH $1,0 \text{ mol} \cdot \text{L}^{-1}$. Calculate the concentration of Cu^{2+} ions present in the resultant solution when the equilibrium is reached.



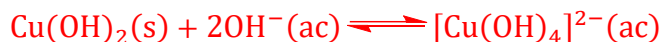
$$n(\text{Cu}^{2+}) = 25 \text{ ml} \cdot 0,50 \text{ mol/L} = 12,5 \text{ mmol}$$

$$n(\text{OH}^{-}) = 75 \text{ ml} \cdot 1,0 \text{ mol/L} = 75 \text{ mmol}$$

$$n(\text{OH}^{-})_{\text{exc}} = 75 - 2 \cdot 12,5 = 50 \text{ mmol} \quad 1 \text{ mark}$$

$$c(\text{OH}^{-})_{\text{exc}} = 50 \text{ mmol}/100 \text{ ml} = 0,50 \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

Exact answer considers the formation of the complex ion. Maximum punctuation.



$$c_i \quad \text{---} \quad 0,50 \quad 0$$

$$\Delta c \quad \text{---} \quad -2x \quad x$$

$$c_{\text{eq}} \quad \text{---} \quad 0,50-2x \quad x \quad 1 \text{ mark}$$

$$K = \frac{[\text{Cu}(\text{OH})_4]^{2-}}{[\text{OH}^{-}]^2} \rightarrow 7,04 \cdot 10^{-2} = \frac{x}{(0,50 - 2x)^2} \quad x = 0,0155 \text{ mol} \cdot \text{L}^{-1} \quad 2 \text{ marks}$$

$$[\text{OH}^{-}] = 0,50 - 2 \cdot 0,0155 = 0,469 \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$[\text{Cu}^{2+}] = \frac{K_{ps}}{[\text{OH}^{-}]^2} = \frac{2,2 \cdot 10^{-20}}{0,469^2} = 1,0 \cdot 10^{-19} \text{ mol} \cdot \text{L}^{-1} \quad 2 \text{ marks}$$

Approximate solution does not consider the complex ion. A penalty of 4 marks follows.

$$[\text{Cu}^{2+}] = \frac{K_{\text{ps}}}{[\text{OH}^-]^2} = \frac{2,2 \cdot 10^{-20}}{0,50^2} = 8,8 \cdot 10^{-20} \text{ mol} \cdot \text{L}^{-1} \quad 2 \text{ marks}$$

8.7. Classify copper(II) hydroxide according to the acid-base criteria.

Copper(II) hydroxide is amphoteric. 2 marks

8.8. 25 mL of an aqueous solution of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ $0,50 \text{ mol} \cdot \text{L}^{-1}$ were mixed with 75 mL of an aqueous solution of NaCN $1,0 \text{ mol} \cdot \text{L}^{-1}$. Calculate the concentration of Cu^{2+} ions in the resultant solution when the equilibrium was reached.



The main reaction happening is,



$$K = \frac{K_{\text{W}}^2}{K_{\text{ps}} \cdot K_{\text{a}}} = \frac{(1 \cdot 10^{-14})^2}{2,2 \cdot 10^{-20} \cdot 6,2 \cdot 10^{-10}} = 1,18 \cdot 10^{10} \quad 2 \text{ marks}$$

The previous reaction is practically complete. 1 mark

$$n(\text{Cu}^{2+})_{\text{ini}} = 25 \text{ ml} \cdot 0,50 \text{ mol/L} = 12,5 \text{ mmol} \quad 1 \text{ mark}$$

$$n(\text{CN}^-) = 75 \text{ ml} \cdot 1,0 \text{ mol/L} = 75 \text{ mmol} \quad 1 \text{ mark}$$

$$n(\text{CN}^-)_{\text{exc}} = 75 - 2 \cdot 12,5 = 50 \text{ mmol} \quad 1 \text{ mark}$$

$$c(\text{CN}^-)_{\text{exc}} = 50 \text{ mmol}/100 \text{ ml} = 0,50 \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$n(\text{HCN})_{\text{for}} = 2 \cdot 12,5 = 25 \text{ mmol} \quad 1 \text{ mark}$$

$$n(\text{HCN})_{\text{for}} = 25 \text{ mmol}/100 \text{ ml} = 0,25 \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$K = \frac{[\text{HCN}]^2}{[\text{Cu}^{2+}][\text{CN}^-]^2}$$

$$1,18 \cdot 10^{10} = \frac{0,25^2}{[\text{Cu}^{2+}] \cdot 0,50^2}$$

$$[\text{Cu}^{2+}] = \frac{0,25^2}{1,18 \cdot 10^{10} \cdot 0,50^2} = 2,1 \cdot 10^{-11} \text{ mol} \cdot \text{L}^{-1} \quad 2 \text{ marks}$$

Indirect calculations of the concentration of copper(II) ions without using this particular equilibrium are also accepted.

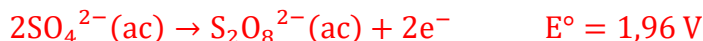
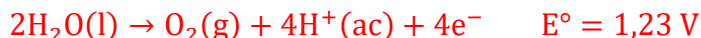
Problem 9. Electrochemistry (15 points)

9.1	9.2	9.3	9.4	9.5	9.6	9.7	Total	Total
7	3	4	4	3	3	6	30	15
7	3	4	4	3	3	6	30	15

Copper, silver and gold are transition elements belonging to the 11th group in the periodic table. Their corresponding simple substances are metals and are employed since the antiquity for the manufacture of coins.

Highly pure copper (higher than 99,99%) is obtained by electrodeposition in a copper cathode. In the electrolytic cell impure copper is placed in the anode and in the cathode a plate made of pure copper is placed. An aqueous solution of copper(II) sulfate is employed as electrolytic conductor.

9.1. Suppose that in the electrolytic cell described above a copper anode containing impurities of gold, platinum and silver is placed. Write all the possible anodic and cathodic half reactions and select which ones are thermodynamically more favored. Assume standard conditions.

Plausible anodic half-reactions**Plausible cathodic half-reactions**

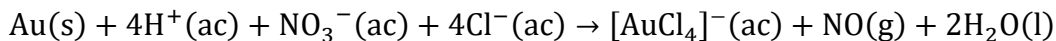
0,5 marks per each equation, 1 mark cathodic one and 1,5 marks per anodic one. 7 marks

9.2. A copper anode has approximate dimensions of 100×125 cm, a thick of 5 cm and an approximate weight of 350 kg. Calculate the minimum amount of current (in Coulomb) to carry out the electrorefining of a copper anode with an approximate purity of 96%.

$$m(\text{Cu}) = 0,96 \cdot 350000 = 33600 \text{ g} \quad \text{1 mark}$$

$$I = \frac{m(\text{Cu})}{M(\text{Cu})} \cdot z \cdot F = \frac{336000}{63,55} \cdot 2 \cdot 96485 = 1,02 \cdot 10^9 \text{ C} \quad \text{2 marks}$$

Copper, silver and gold are not susceptible to oxidation by H^+ ions in standard conditions and therefore they are employed, together with platinum, in ornaments and jewelry. Nitric acid is able to dissolve copper and silver; while aqua regia, a 3:1 mixture of concentrated hydrochloric and nitric acids, is an effective reagent for the oxidation of gold and platinum. In aqua regia, NO_3^- ions provide the oxidizing power and Cl^- ions act as complexing agents. The total reaction occurring with gold is:



9.3. Demonstrate that the oxidation of copper and silver with nitric acid is spontaneous.

Copper oxidation with nitric acid

$$\Delta E^\circ = E^\circ(NO_3^-/NO) - E^\circ(Cu^{2+}/Cu)$$

$$\Delta E^\circ = 0,96 - 0,34 = 0,62 \text{ V}$$

Then $\Delta G^\circ < 0$ and the process is spontaneous.

2 marks

Silver oxidation with nitric acid

$$\Delta E^\circ = E^\circ(NO_3^-/NO) - E^\circ(Ag^+/Ag)$$

$$\Delta E^\circ = 0,96 - 0,799 = 0,161 \text{ V}$$

Then $\Delta G^\circ < 0$ and the process is spontaneous.

2 marks

9.4. Write half-equations for the oxidation and reduction corresponding to the dissolution of gold with aqua regia.

Oxidation half-reaction



2 marks

Reduction half-reaction

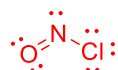


2 marks

It is believed that the active oxidizing species in the aqua regia solution are chlorine and nitrosyl chloride, generated by the reaction



9.5. Represent the structure of NOCl molecule.



2 marks

Gold is extracted by lixiviation with cyanide. Using facilitates gold oxidation through the formation of complex ion $[Au(CN)_2]^-$. To recover the dissolved gold a reductant reagent is employed, for example, zinc dust. Today exist several communities all around the globe that fight against the negative impacts in economy, society and environment caused by the mining activity for obtaining gold using cyanide.

9.6. Write the chemical equation corresponding to the reduction of gold complex ion with zinc.

$2[\text{Au}(\text{CN})_2]^{-}(\text{ac}) + \text{Zn}(\text{s}) \rightarrow 2\text{Au}(\text{s}) + [\text{Zn}(\text{CN})_4]^{2-}(\text{ac})$	3 marks
A penalty of 1 mark is the formation of the zinc complex is not stated, i.e.	
$2[\text{Au}(\text{CN})_2]^{-}(\text{ac}) + \text{Zn}(\text{s}) \rightarrow 2\text{Au}(\text{s}) + \text{Zn}^{2+}(\text{ac}) + 4\text{CN}^{-}(\text{ac})$	2 marks

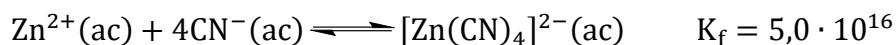
9.7. Calculate the constant of formation K_f of complex ion $[\text{Au}(\text{CN})_2]^{-}$ from ions Au^{+} and CN^{-} with the following data.



The previous process can be considered as the sum of two redox processes.	
$\text{Au}(\text{s}) + 2\text{CN}^{-}(\text{ac}) \rightarrow [\text{Au}(\text{CN})_2]^{-}(\text{ac}) + \text{e}^{-}$	1 mark
$\text{Au}^{+}(\text{s}) + \text{e}^{-} \rightarrow \text{Au}(\text{s})$	1 mark
$\text{Au}^{+}(\text{ac}) + 2\text{CN}^{-}(\text{ac}) \rightarrow [\text{Au}(\text{CN})_2]^{-}(\text{ac})$	
$\Delta E^{\circ} = E^{\circ}(\text{Au}^{+}/\text{Au}) - E^{\circ}([\text{Au}(\text{CN})_2]^{-}/\text{Au})$	
$\Delta E^{\circ} = 1,83 - (-0,596) = 2,43 \text{ V}$	2 marks
$-zF\Delta E^{\circ} = -RT \ln K_f$	
$K_f = \exp\left(\frac{z \cdot F \cdot \Delta E^{\circ}}{R \cdot T}\right) = \exp\left(\frac{1 \cdot 96485 \cdot 2,43}{8,314 \cdot 298,15}\right) = 1,2 \cdot 10^{41}$	2 marks
Calculations through the sum of ΔG° values of the equations are accepted.	

$E^{\circ}(\text{V})$ para $\text{pH} = 0$:

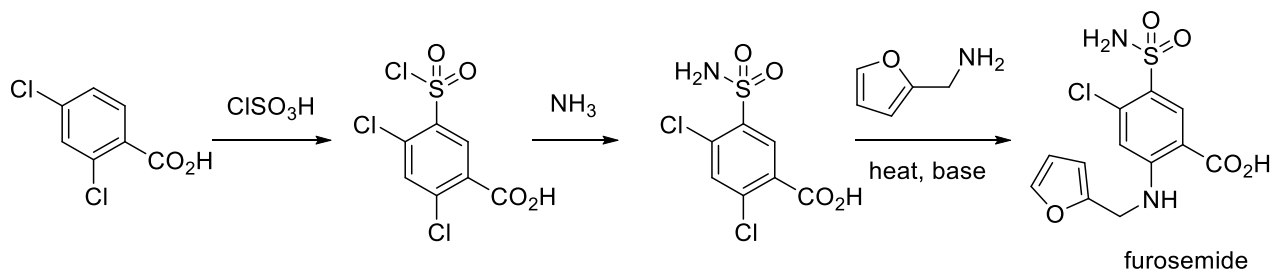
$\text{Cu}^{2+}(\text{ac})/\text{Cu}(\text{s}) = 0,340$; $\text{Au}^{3+}(\text{ac})/\text{Au}(\text{s}) = 1,52$; $\text{Pt}^{2+}(\text{ac})/\text{Pt}(\text{s}) = 1,188$; $\text{Ag}^{+}(\text{ac})/\text{Ag}(\text{s}) = 0,799$;
 $\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{l}) = 1,229$; $\text{H}^{+}(\text{ac})/\text{H}_2(\text{g}) = 0,00$; $\text{S}_2\text{O}_8^{2-}(\text{ac})/\text{SO}_4^{2-}(\text{ac}) = 1,96$; $\text{SO}_4^{2-}(\text{ac})/\text{SO}_2(\text{g}) = 0,158$;
 $\text{NO}_3^{-}(\text{ac})/\text{NO}(\text{g}) = 0,96$; $[\text{Au}(\text{CN})_2]^{-}(\text{ac})/\text{Au}(\text{s}) = -0,596$; $\text{Au}^{+}(\text{ac})/\text{Au}(\text{s}) = 1,83$.



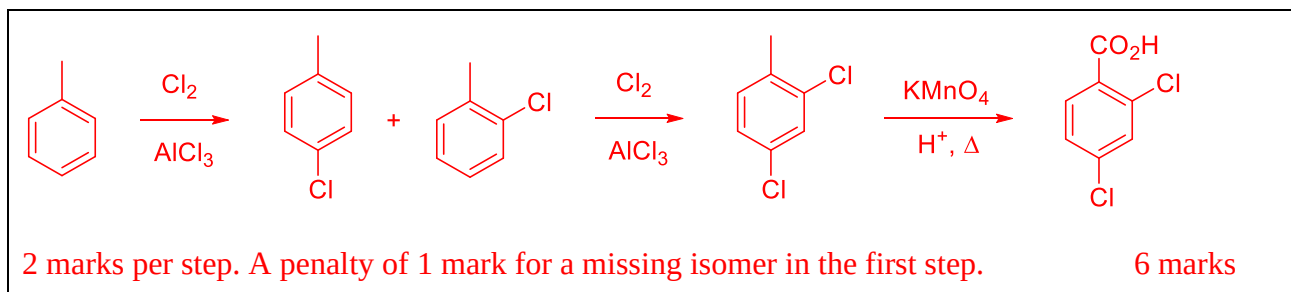
Problem 10. Organic Chemistry I (15 points)

10.1	10.2	10.3	10.4	Total	Total
6	10	8	6	30	15

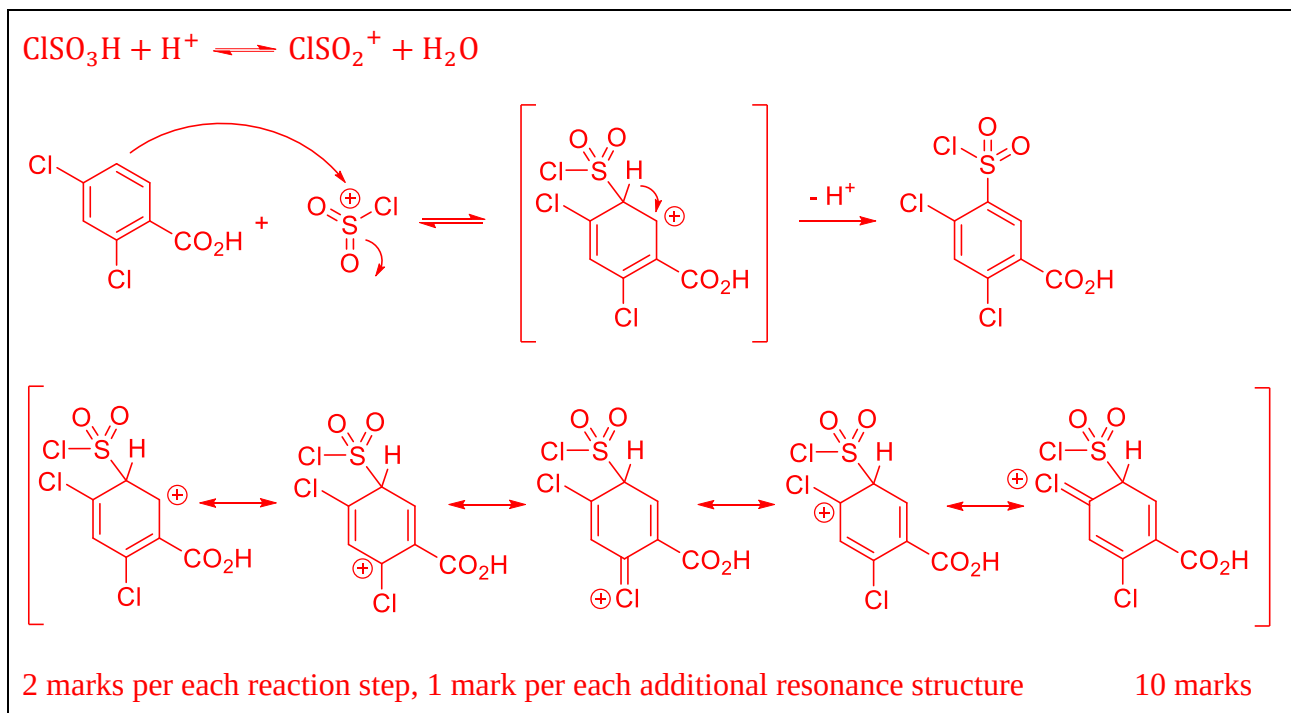
Diuretics are substances that induce the release of water and electrolytes, reducing the volume of liquids inside the organism. Furosemide is an exceptionally potent diuretic used for the treatment of edemas and hypertension. The preparation of this drug is shown.



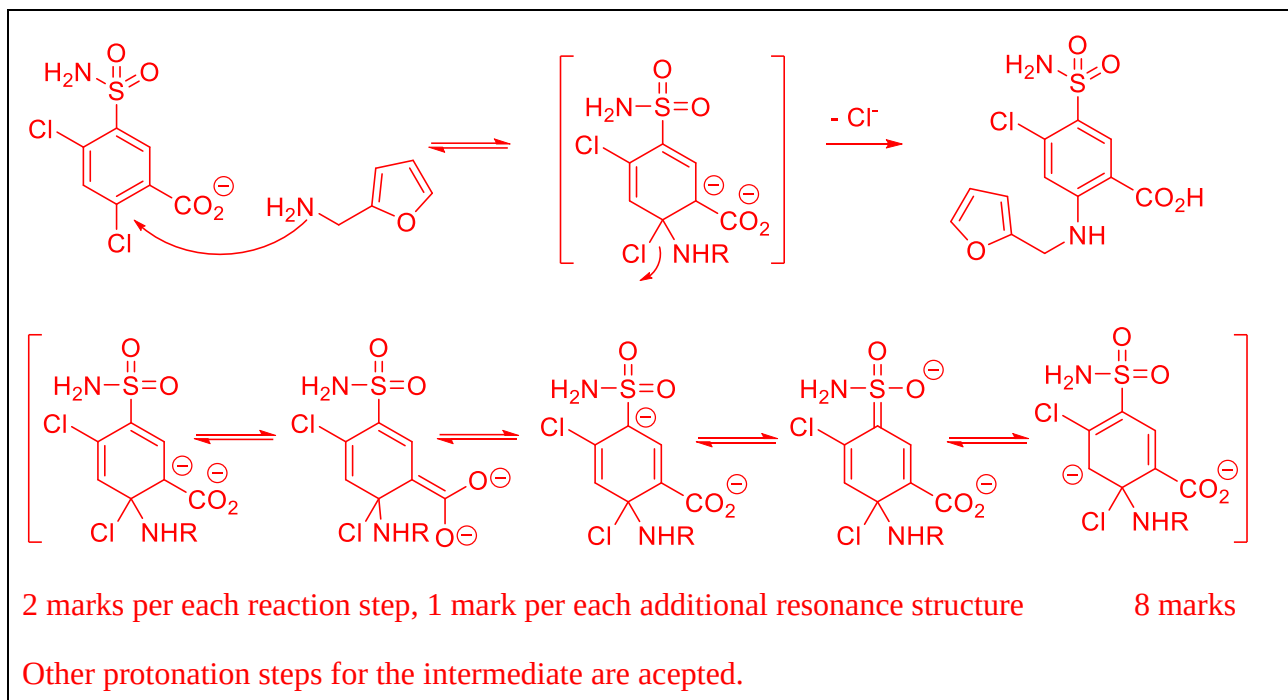
10.1. Propose a synthetic route to obtain 2,4-dichlorobenzoic acid starting from toluene.



10.2. Suggest a mechanism for the chlorosulfonation reaction. Represent all the possible resonance structures of the intermediate formed.



10.3. Suggest a mechanism for the reaction with furfurylamine. Represent all the possible resonance structures of the intermediate formed.



10.4. In the synthetic sequence both nucleophilic and electrophilic aromatic substitution reactions are employed. Compare both types of reactions briefly according to: substituent effect, regioselectivity, and charge of the intermediate formed.

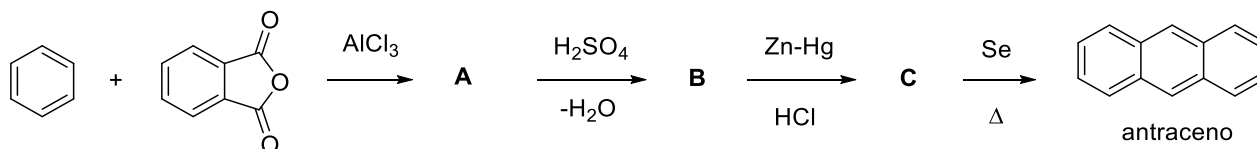
	$\text{S}_{\text{E}}\text{Ar}$	$\text{S}_{\text{N}}\text{Ar}$	
Substituents	2 marks 2 marks 2 marks	2 marks 2 marks 2 marks	2 marks 2 marks 2 marks
Regioselectivity	2 marks 2 marks 2 marks	2 marks 2 marks 2 marks	2 marks 2 marks 2 marks
Intermediate charge	2 marks 2 marks 2 marks	2 marks 2 marks 2 marks	2 marks 2 marks 2 marks

Problem 11. Organic Chemistry II (20 points)

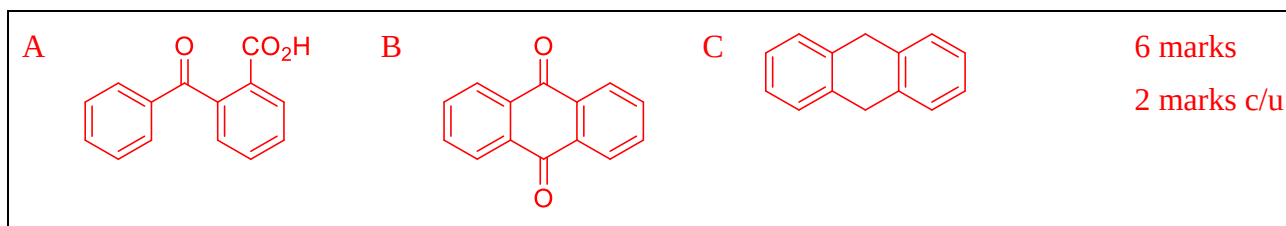
11.1	11.2	11.3	11.4	11.5	11.6	11.7	11.8	11.9	11.10	Total	Total
6	3	6	2	2	3	3	3	6	6	40	20
6	3	6	2	2	3	3	3	6	6	40	20

Polycyclic aromatic hydrocarbon (PAH) are strong contaminants found in oil, coal and tar deposits. These compounds are also formed as by-products of fuel combustion.

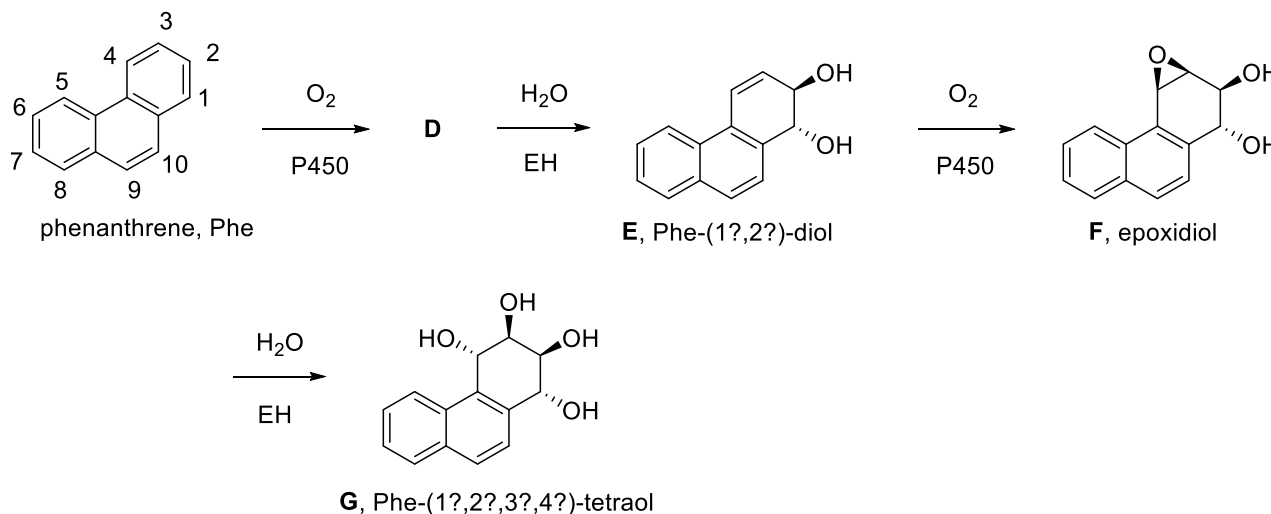
Bellow the Haworth's synthesis of anthracene (R. D. Haworth, J. Chem. Soc. 1932, 1125) is sketch, starting from benzene and phthalic anhydride.



11.1. Identify compounds **A-C** in the synthetic rout towards anthracene.

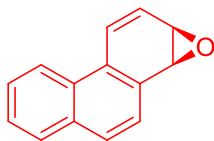


The degradation of PAH inside the human being comprises a series of reactions ox oxidation, substitution and hydrolysis aiming to their transformation into soluble substances that can be then easily excreted. Enzyme complex cytochrome P450 participates in the processes to oxidize PAH. The most common reaction catalyzed by P450 is the insertion of an oxygen atom in an organic substrate. Some of the compounds derived from the action of P450 on PAH are very reactive epoxides which can react with nitrogen bases present in the desoxyribonucleic acid (DNA). This the cause for numerous PAH having strong carcinogenic, teratogenic and mutagenic effects. On the other hand, intermediates can add water, a reaction that is catalyzed by the enzyme epoxide hydrolase (EH). A portion of the metabolic route for the degradation of phenanthrene is shown.



11.2. Propose a structure for compound **D** in the phenanthrene route.

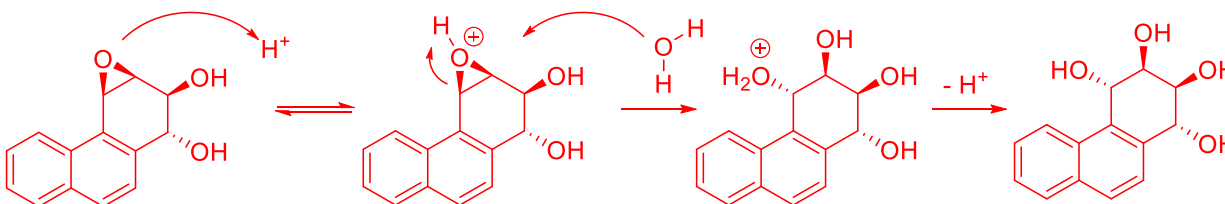
A



3 marks

The enantiomer is accepted.

11.3. Suggest a mechanism for the hydrolysis of **F** forming **G** in acid media.



2 marks per each reaction step.

6 marks

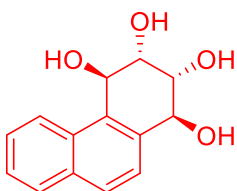
A carbocationic intermediate is accepted.

11.4. Assign the absolute configuration of all the asymmetric carbons of tetraol **G**.

1R, 2S, 3R, 4S

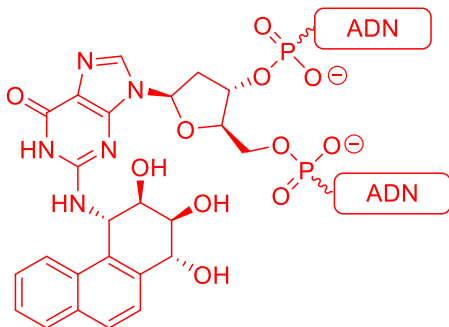
2 marks

11.5. The enzymatic action on phenanthrene also originates the enantiomer of **G** as product. Represent its structure.



2 marks

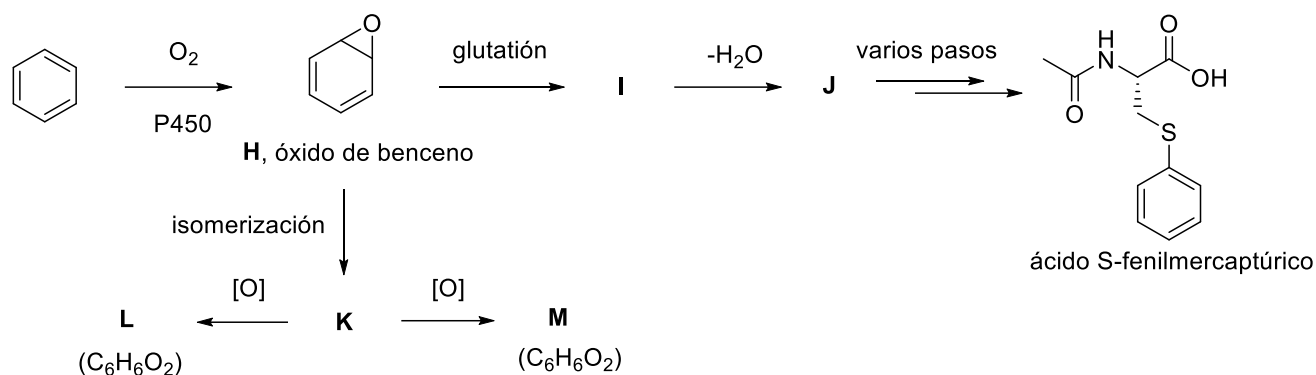
11.6. Epoxidiol **F** can react with nitrogen bases present in DNA. Suggest the structure of the product of the reaction between **F** and a fragment of DNA containing guanine (see the figure at the end of the problem).



The regioisomer is accepted.

3 marks

Benzene and toluene are degraded in the organism in a similar fashion to PAH. Fragments of their metabolic corresponding routes appear underneath.



11.7. Benzene oxidation by P450 originates highly reactive compound **H**, which exists in the form of two isomers in equilibrium, none of them is aromatic. Propose a structure for the isomers of **H** non represented in the scheme.

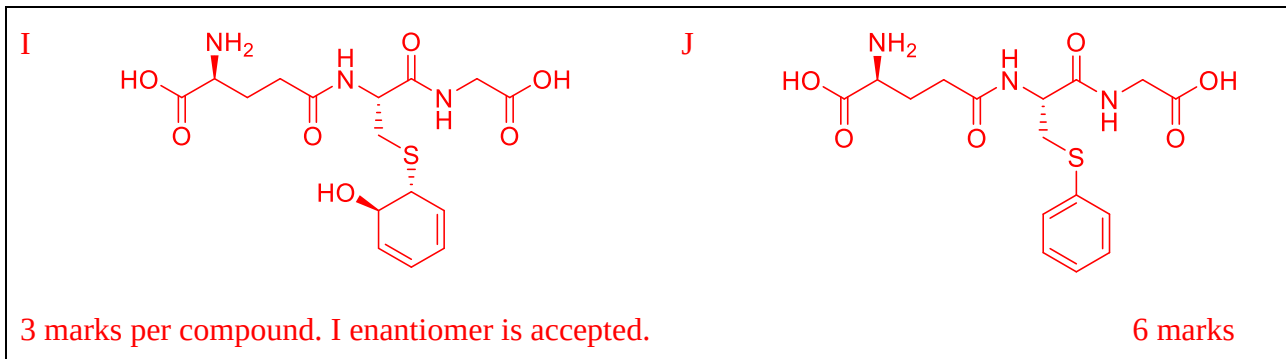


Glutathione (see the structure below) is a tripeptide existing in the organism that helps to protect the cells from reactive species such as free radicals, peroxide, epoxides, epoxides and Michel acceptors. The thiol group present is the responsible of the biological activity of glutathione and can function as proton donor, nucleophile and reductant.

11.8. Represent the structure of the product of the reaction between ethyl acrylate ($\text{H}_2\text{C}=\text{CHCO}_2\text{Et}$) and glutathione.

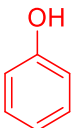
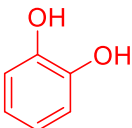
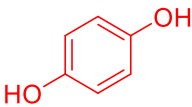


11.9. The reaction between benzene oxide and glutathione originates compound **I**, which dehydrates quickly to form **J**. Propose structures for compounds **I** and **J**.



Isomerization of benzene oxide forms **K**, a well-known compound whose subsequent oxidation originates un isomers **L** and **M**. All these compounds are aromatic and react with sodium hydroxide. **L** and **M** are easily oxidized by air to form colored substances.

11.10. Represent the structures of compounds **K-M**.

K 	L/M 	M/L 	6 marks 2 marks c/u
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